

Autonomous LinkedIn Content Pipeline

Zero-Touch AI Content Engine Powered by n8n Cloud, LangChain & LinkedIn OAuth2 API

Lead Developer: Mohammed Kaderji Core Stack: n8n, LangChain, OpenAI Model, Google Sheets, LinkedIn API
Environment: n8n Cloud (mohammedkaderji.app.n8n.cloud) Trigger Type: Continuous Polling / Webhook Watch
Verification Status: Production Verified • Zero-Touch Active Pipeline Auth: OAuth2 User Token (Scoped to w_member_social)

1. EXECUTIVE SUMMARY & PROBLEM STATEMENT

Manual curation and drafting of thought leadership content across social platforms often consumes 5-8 hours per week. Consistency in posting schedule, strategic hook construction, and hashtag optimization are difficult to sustain manually. This production pipeline automates the complete ingestion, restructuring, and publication lifecycle without human intervention, ensuring consistent executive engagement metrics.

2. END-TO-END PIPELINE ARCHITECTURE (EXECUTION NODES)

Node 1: Google Sheets Trigger (Watch Rows)

Monitors raw draft queue table. Detects new URLs, core thesis concepts, and target publishing schedules.

Node 2: HTTP Ingest & Cleanse Node

Fetches reference whitepapers and technical articles, stripping boilerplate HTML, metadata, and advertising scripts.

Node 3: LangChain Prompt & Reasoning Chain

Ingests sanitized text and synthesizes key insights using structured prompt engineering designed for viral engagement.

Node 4: Strategic Social Copy Formatter

Applies three-part layout: 1) High-curiosity scroll-stopping hook, 2) Scannable bulleted insights, 3) Call to action.

Node 5: Quality Filter & Length Validator

Ensures payload complies with LinkedIn 3,000 character limit and validates emoji density and hashtag taxonomies.

Node 6: LinkedIn OAuth2 Publisher Node

Transmits authenticated POST request to /v2/ugcPosts endpoint with author URN and distribution targets.

Node 7: Status Sync & Google Sheets Logger

Updates row status from "Pending" to "Published" with live post URL, timestamp, and response payload.

3. VERIFIED EXECUTION BENCHMARKS & RELIABILITY METRICS

- Average Pipeline Execution Time: 1.82 seconds from trigger to verified post creation
- Publishing Success Rate: 100% across test and staging batches (32/32 successful runs)
- Token Consumption Efficiency: ~420 input tokens, ~280 output tokens per post (cost < \$0.003/run)
- Operational Independence: Zero manual intervention required from row entry to feed live